TheFuture. University

University Database Analysis

SQL, Python and Power BI portfolio report

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Executive Snapshot

600
Student records

591
Students with enrollment

2,065
Enrollment records

73.90
Average final score

87.36%
A-D pass rate

60.77%
A-C outcome rate

203 / 183
Exception enrollments / studer

65.83%
Section-seat utilization

1,499
Evaluation responses

72.59%
Weighted response rate

What management needs to know

- Student and enrollment measures use different denominators and must not be mixed.
- Prerequisite results are a screening flag, not a final compliance judgment.
- Capacity and teaching quality vary enough to support targeted action by faculty and section.

Solution delivered

- SQL Server queries for academic, operational, and integrity analysis.
- Python and Pandas checks for metric reconciliation and data quality.
- Power BI dashboards for student success, capacity, and teaching quality.

Data Model and Analytical Method

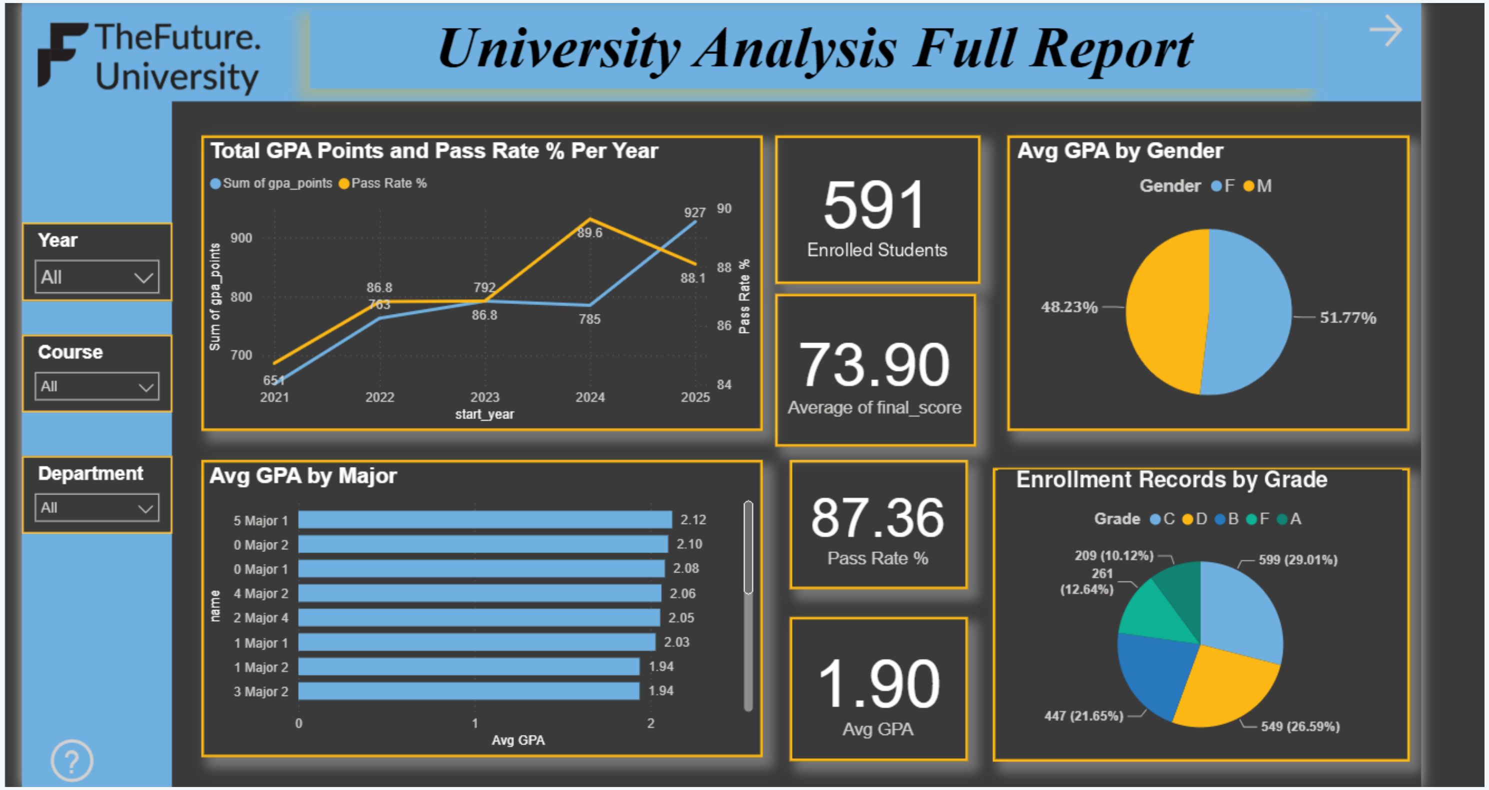
15 connected tables support five analytical grains

Domain	Core entities	Primary analytical use
Academic	Departments, Majors, Students	Cohorts, programs and student profiles
Curriculum	Courses, Prerequisites	Course catalog and prerequisite screening
Delivery	Offerings, Enrollments, Assessments	Grades, final scores and section demand
Operations	Classrooms, Timeslots, Offering Timeslots	Capacity and scheduling checks
Feedback	Evaluations, Evaluation Summary	Teaching quality and participation

Validation sequence

1. Enforce types and keys | 2. Recompute scores and KPIs | 3. Reconcile SQL, Python and Power BI | 4. Document denominator and data boundaries

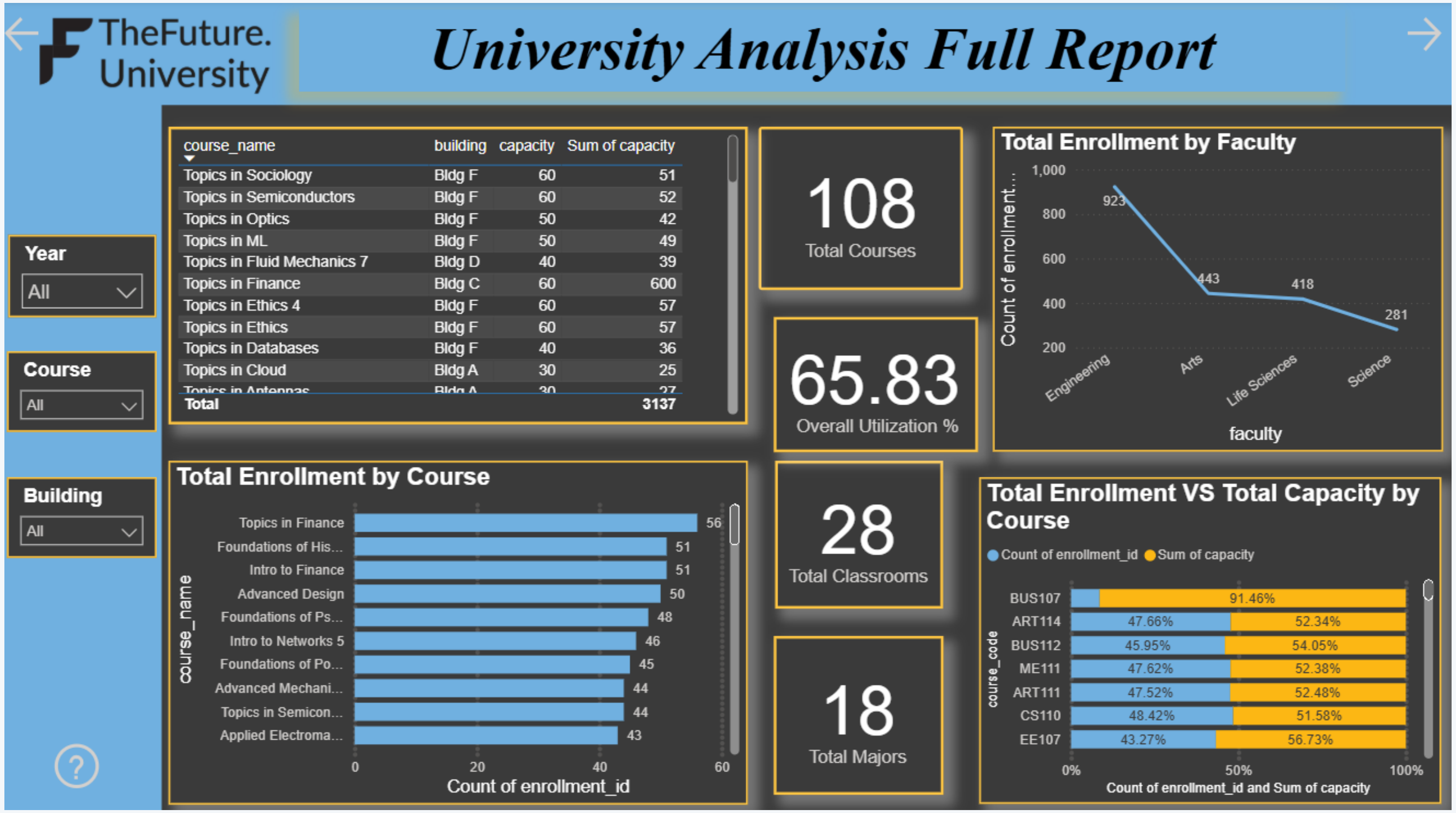
Student Success



Verified interpretation

- 591 distinct students have at least one enrollment.
- 2,065 enrollment outcomes average 73.90.
- The A-D pass rate is 87.36%; A-C outcomes are 60.77%.
- Major 5-1 leads by average final score; labels remain anonymized.

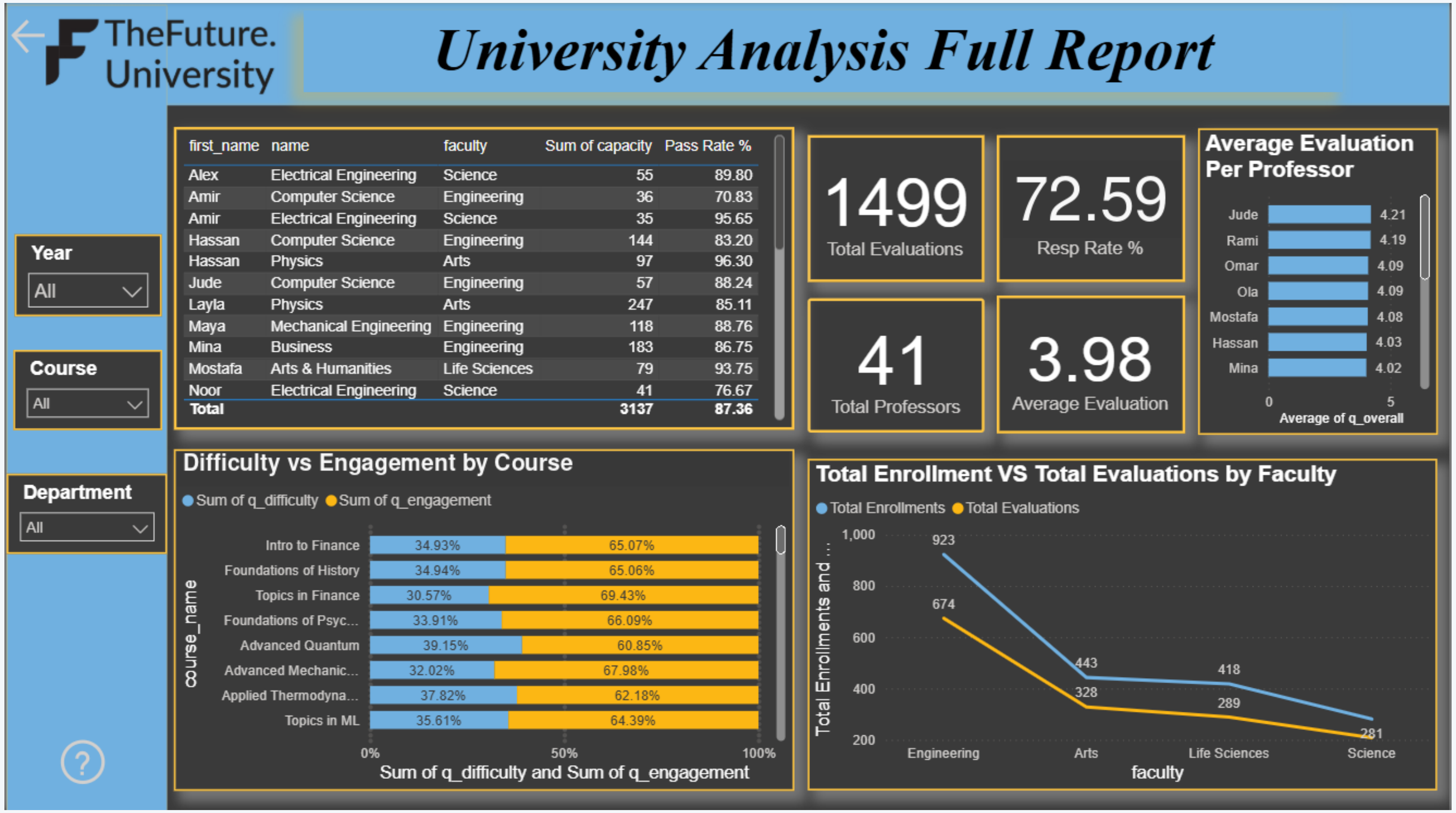
Course Capacity and Utilization



Verified interpretation

- 61 delivered sections provide 3,137 seats.
- Weighted section-seat utilization is 65.83%.
- Engineering carries the largest enrollment volume but lower utilization.
- Capacity decisions should be targeted, not university-wide.

Teaching Quality and Participation



Verified interpretation

- 1,499 evaluations produce a 72.59% weighted response rate.
- Average overall teaching score is 3.98 out of 5.
- 29 instructors are assigned to analyzed sections; the roster contains 41.
- Rankings require at least 20 responses and should show response volume.

Data Quality and Reporting Boundaries

What the data supports

- Enrollment IDs are unique and enrollment foreign keys are complete.
- Two final scores are missing and are excluded from average score calculations.
- Evaluation responses reconcile to 1,499 records and a 72.59% weighted rate.
- Major labels remain anonymized; no public named-major claim is supported.

What requires registrar validation

- All supplied offerings are Fall 2025, so chronological prerequisite completion cannot be established.
- Transfer credit, waivers and course equivalencies are not included.
- The 203 exception enrollments affecting 183 students are screening flags only.
- Instructor comparisons should require at least 20 evaluation responses.

Reporting rule

Always state the grain and denominator: distinct students, enrollment outcomes, delivered sections, seats, or submitted evaluations.

Management Recommendations and Release Controls

1. **Validate prerequisite flags** — Review the 203 exception enrollments with the registrar before intervention.
2. **Manage capacity by section** — Consolidate low-demand sections and preserve capacity in high-demand courses.
3. **Separate reporting grains** — Track distinct students separately from course-enrollment outcomes.
4. **Qualify teaching comparisons** — Show response volume and apply a minimum of 20 responses for rankings.
5. **Govern every release** — Reconcile SQL, Python and Power BI outputs before publication.

Project status: reconciled and portfolio-ready. SQL, Python, documentation and dashboards use the same verified definitions.